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Foundations of Computer Systems

Data Representation

- Data and its types.
- Numeric data representation.
- Text data representation.
- Image data representation.

Data

- Data

Collection of raw, unorganized facts.

Consists of numbers, text, symbols or just description of things.

- Information

Processed data is called information.

Data and Information

- Data
  - Unprocessed
  - Unorganized
  - Non-structured
  - eg:- Chaithra, 83, Science, Maths, Rithika, 95
- Information
  - Processed
  - Organized
  - Structured

Student Name	Course	Marks Obtained
Chaithra	Maths	95
Rithika	Science	83

Types of Data

- Numeric
- Character String (Text)
- Audio
- Images
- Videos

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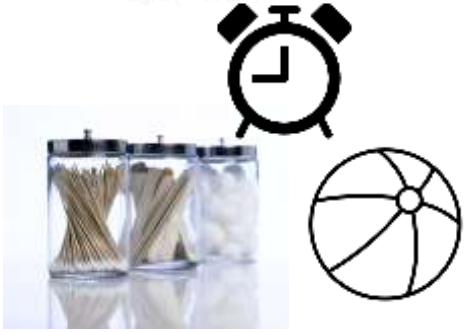
Data Representation

- Data is stored, processed and transmitted as sequence of bits (Binary Digits), ie 0's and 1's.
- Even though human interpret text and pictures; computers interpret them as bit patterns.



123abc

100000100000
100001110010
101011110011
110011010110

Numeric Data Representation

- The numbers are encoded to their binary equivalent through number conversion methods.
- Separate conversion methods for representing.
  - Integers
  - Fractions numbers

Text Data Representation

- Represented by means of a code
  - **ASCII**
  - **Unicode**
- Each of the different symbols in the text is assigned a unique bit pattern.
- The text is then represented as a long string of bits.

Audio Representation

- The **amplitude** of the sound wave is recorded at regular intervals through **sampling**.
- The sampled data is then encoded to binary sequence.

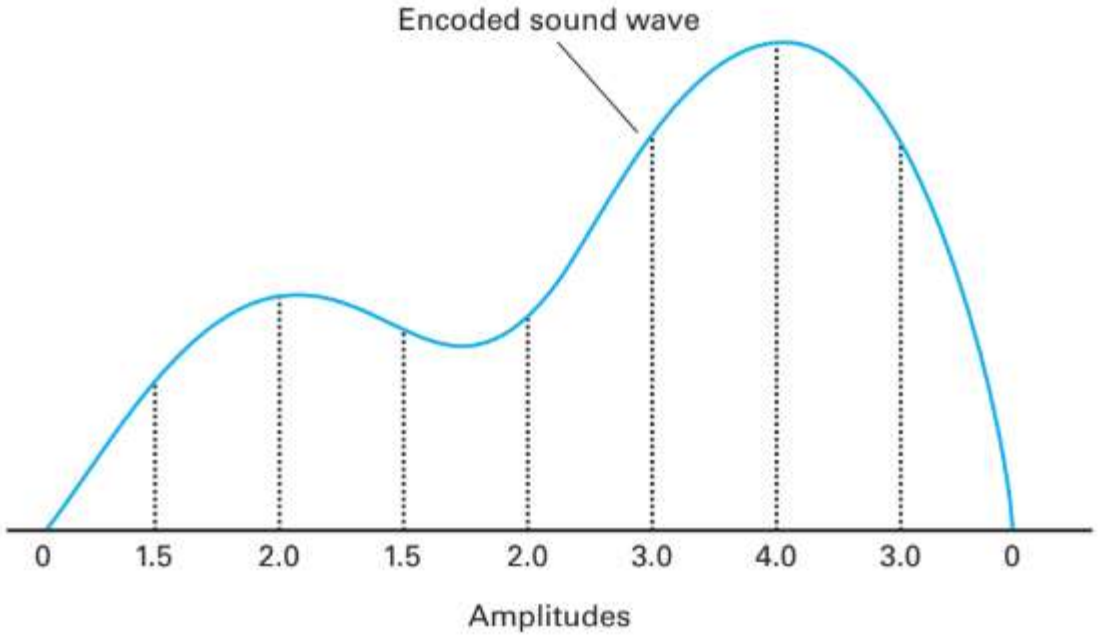


Image & Video Representation

- Image is interpreted as a collection of dots, each of which is called a **pixel** (picture element).
- Each pixel is then encoded and the entire image is represented as a collection of these encoded pixels.
- A video can be considered as a collection of images and can be encoded almost similar to image encoding.

Takeaways

- What is data?
- Data vs Information.
- Different types of data.
- How each type is represented?

References

[1] Fundamentals Computers, V Rajaraman, Neeharika Adabala, Phi Learning Pvt. Ltd

[2] J. Glenn Brookshear,"Computer Science: An Overview", Addision-Wesley, Twelth Edition, 2014

[3] Computer Fundamentals, Anita Goel, Pearson

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Foundations of Computer Systems

Objectives

- Different types of number systems.
- Unsigned number representation.

Numeric Data

- Consists of numbers that can be used in arithmetic operations.
- Two types are:
    - Unsigned
    - Signed

Numeric Data Representation

- **Unsigned**
  - Binary conversion
- **Signed**
  - Sign and Magnitude
  - One's Complement
  - Two's Complement

Number Systems

- Defines how a number can be represented using distinct symbols or digits.
- Base in a number system is the total number of distinct digits in the system.
- The maximum value of a single digit in a number system is one less than its base.

Types of Number System

- Decimal
- Binary
- Octal
- Hexadecimal

Number System	Base	Digits
Decimal	10	0 1 2 3 4 5 6 7 8 9
Binary	2	0 1
Octal	8	0 1 2 3 4 5 6 7
Hexadecimal	16	0 1 2 3 4 5 6 7 8 9 A B C D E F

Number Notation

- Decimal number
  - Example: $(29)_{10}$ commonly represented as 29
- Binary number
  - Example: $(1100)_2$
- Octal number
  - Example: $(35)_8$
- Hexadecimal number
  - Example: $(1D)_{16}$

Unsigned Numeric Data Representation

- Any number system can be represented to its binary equivalent by the binary conversion technique.
- Decimal (Base 10) to binary conversion.
- Octal to binary conversion.
- Hexadecimal to binary conversion.

Decimal to Binary Conversion

- A decimal number can be converted into its equivalent in base 2 using the following procedure:

  - Step 1: Perform the integer division of the decimal number by 2 and record the remainder.
  - Step 2: Replace the decimal number with the result of the division in step 1. Repeat step 1, until a zero result is found.
  - Step 3: The binary number is formed by reading the remainders in reversed order.

Decimal (Base 10) to Binary Conversion

$(160)_{10}$

2	160	
2	80	0
2	40	0
2	20	0
2	10	0
2	5	0
2	2	1
2	1	0
	0	1

$(160)_{10} = (10100000)_2$

Floating Point Decimal Number to Binary Conversion

$(169.9)_{10}$

Integer Part = 169

2	169	
2	84	1
2	42	0
2	21	0
2	10	1
2	5	0
2	2	1
2	1	0
	0	1

Fractional Part = 0.9

	.9 x 2 = 1.8
1	.8 x 2 = 1.6
1	.6 x 2 = 1.2
1	.2 x 2 = 0.4
0	.4 x 2 = 0.8
0	.8 x 2 = 1.6
1	.6 x 2 = 1.2
1	.2 x 2 = 0.4
0	.4 x 2 = 0.8

169 = 10101001

0.9 = 1110011

$(169.9)_{10}$ =

$(10101001.1110011)_2$

Takeaways

- What is numeric data?
- What are the different types of number systems?
- Unsigned numeric data representation techniques
  - Unsigned integer
  - Unsigned
  - Floating point number

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Foundations of Computer Systems

Objectives

- Signed number representation.
- 1's complement.
- 2's complement.

Signed Numeric Data

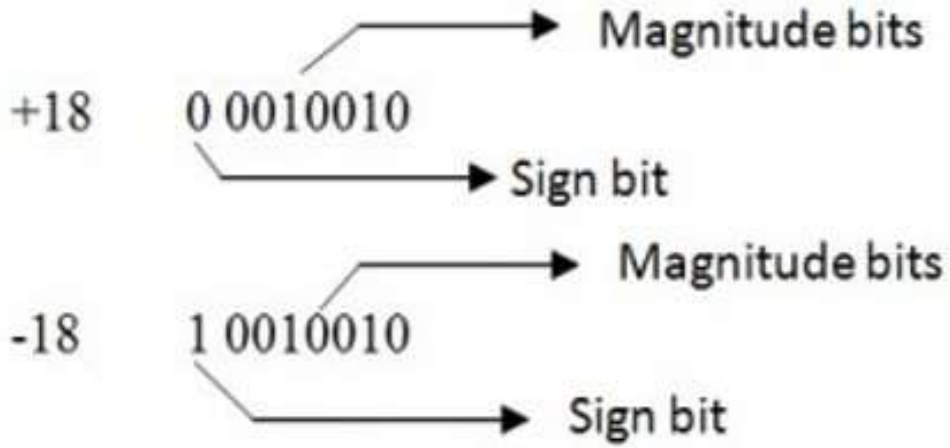
- Can represents zero, positive, negative integers.
- Contain sign flag.
- In binary number system signed numbers are used to encode negative numbers.

Signed Numeric Data Representation

- Signed Magnitude Method
- 1's Complement Method
- 2's Complement Method

Signed Magnitude Method

- One of the simplest representation.
- MSB (Most Significant Bit) represents the sign
  - 0 represents +ve number
  - 1 represents -ve number
- Remaining bits represent magnitude.



Signed Magnitude Method

- A 'n' bit binary number can represent numbers from

  – $(2^{n-1}-1)$ to $+(2^{n-1}-1)$

Base 10 Number	Signed Magnitude Representation
-3	111
-2	110
-1	101
0	000
1	001
2	010
3	011

-1

Binary equivalent of 1 in two bit : 01

Including MSB bit: **1** 01

111

Decimal equivalent of 11 is: 3

Number represented is: **-3**

1’s Complement Method

- The positive numbers are represented just like the normal binary representation.
- The negative numbers are represented using the 1’s complemen positive counterpart.

Base 10 Number	1's Complement Representation
+5	0101
-5	1010
+14	01110
-14	10001

1's Complement Method

- A 'n' bit binary number can represent numbers from

$-(2^{n-1}-1)$ to $+(2^{n-1}-1)$

Base 10 Number	1's Complement Representation
-3	100
-2	101
-1	110
0	000
1	001
2	010
3	011

100

1's complement of 100 is: 011

Decimal equivalent of 011: **3**

Obtaining 2's complement for a binary number

- Flip all bits to obtain its 1's complement equivalent.

Binary number		1's Complement
0	→	1
1	→	0
1011	→	0100
0011	→	1100
11100010 00001111	→	00011101 11110000

2's Complement Method

- The positive numbers are represented just like the normal binary representation.
- The negative numbers are represented using the 2's complement representation of its positive counterpart.

Base 10 Number	2's Complement Representation
+5	0101
-5	1011
+14	01110
-14	10010

2's Complement Method

- An 'n' bit binary number can represent numbers from

  $-(2^{n-1})$ to $+ (2^{n-1}-1)$.

Base 10 Number	2's Complement Representation
-4	100
-3	101
-2	110
-1	111
0	000
1	001
2	010
3	011

101

2's complement of 101 is: 011

Decimal equivalent of 011: **3**

Obtaining 2's complement for a binary number

- First find the 1's complement.
- Then add 1 to it to obtain the 2's complement equivalent.

Binary number : 11000010

↓

1's Complement : 001111101

↓ Add 1

2's Complement : 001111110

Takeaways

- What are signed numbers.
- Way of representing signed numbers.
- Signed magnitude representation.
- 1's Complement.
- 2's Complement.

References

[1] Fundamentals Computers, V Rajaraman, Neeharika Adabala, Phi Learning Pvt. Ltd

[2] J. Glenn Brookshear,"Computer Science: An Overview", Addision-Wesley, Twelth Edition, 2014

[3] Computer Fundamentals, Anita Goel, Pearson

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Foundations of Computer Systems

Objectives

- Types of text data.
- ASCII representation.
- Unicode representation.

Text Data and its Representation

- Text Data
  - Alpha-numeric data
  - Special characters
- Mainly represented using
  - ASCII
  - Unicode
- Each symbols are assigned with a special code.
- The code is then converted to binary.
- Finally, the entire text is represented as a long string of bits.

ASCII-Facts

- **ASCII** (American Standard for Information Interchange) is a coding system for representing characters.
- It uses 8 bits to represent a character.
- The MSB is set as the parity bit. So it can represent only 128 characters.
- Extended ASCII discards the parity bit and it can represent 256 characters.

ASCII Codes

Symbol	ASCII	Hex	Symbol	ASCII	Hex	Symbol	ASCII	Hex
line feed	00001010	0A	>	00111110	3E	^	01011110	5E
carriage return	00001011	0B	?	00111111	3F	_	01011111	5F
space	00100000	20	@	01000000	40	`	01100000	60
!	00100001	21	A	01000001	41	a	01100001	61
"	00100010	22	B	01000010	42	b	01100010	62
#	00100011	23	C	01000011	43	c	01100011	63
$	00100100	24	D	01000100	44	d	01100100	64
%	00100101	25	E	01000101	45	e	01100101	65
&	00100110	26	F	01000110	46	f	01100110	66
'	00100111	27	G	01000111	47	g	01100111	67
(	00101000	28	H	01001000	48	h	01101000	68
)	00101001	29	I	01001001	49	i	01101001	69
*	00101010	2A	J	01001010	4A	j	01101010	6A
+	00101011	2B	K	01001011	4B	k	01101011	6B
,	00101100	2C	L	01001100	4C	l	01101100	6C
-	00101101	2D	M	01001101	4D	m	01101101	6D
.	00101110	2E	N	01001110	4E	n	01101110	6E
/	00111111	2F	O	01001111	4F	o	01101111	6F
0	00110000	30	P	01010000	50	p	01110000	70
1	00110001	31	Q	01010001	51	q	01110001	71
2	00110010	32	R	01010010	52	r	01110010	72
3	00110011	33	S	01010011	53	s	01110011	73
4	00110100	34	T	01010100	54	t	01110100	74
5	00110101	35	U	01010101	55	u	01110101	75
6	00110110	36	V	01010110	56	v	01110110	76
7	00110111	37	W	01010111	57	w	01110111	77
8	00111000	38	X	01011000	58	x	01111000	78
9	00111001	39	Y	01011001	59	y	01111001	79
:	00111010	3A	Z	01011010	5A	z	01111010	7A
;	00111011	3B	[	01011011	5B	{	01111011	7B
<	00111100	3C	\	01011100	5C	|	01111100	7C
=	00111101	3D	]	01011101	5D	}	01111101	7D

ASCII Table

- ASCII value of number 0 :- 48
  - Numeric range → 48-57
- ASCII value of capital letter A :- 65
  - Alpha range (Uppercase Alphabets) → 65-90
- ASCII value of small letter a :- 97
  - Alpha range (Lowercase Alphabets) → 97-122
- ASCII value of space :- 32

ASCII Table

- ASCII value of number 0 :- 48
  - Numeric range → 48-57
- ASCII value of capital letter A :- 65
  - Alpha range (Uppercase Alphabets) → 65-90
- ASCII value of small letter a :- 97
  - Alpha range (Lowercase Alphabets) → 97-122
- ASCII value of space :- 32

Text Representation using ASCII

- Eg Computer Essentials.
- Step 1: Convert each character to its ASCII Decimal value using the ASCII table.

  Like C = 67, o = 111, m = 109, p = 112, …etc

Text Data		Text Representation using ASCII
Computer Essentials	⇨	67 111 109 112 117 116 101 114 32 69 115 115 101 110 116 105 97 108 115

Text Conversion using ASCII

- Step 2 : Convert each decimal value into its 8-bit equivalent binary.
  - 77 = 01001101 121= 01111001 32= 00100000 ….

Text Representation using ASCII

67 111 109 112 117 116 101 114
32 69 115 115 101 110 116 105 97
108 115

⇨

Binary Representation

01000011 01101111 01101101
01110000 01110101 01110100
01100101 01110010 00100000
01000101 01110011 01110011
01100101 01101110 01110100
01101001 01100001 01101100
01110011

Unicode - Facts

- ASCII : The first widespread encoding scheme.
- It's limited to only 128 character definitions.
- The rest of the world wants the same encoding scheme for their characters too.
- Later, a new character encoding scheme was needed which is when the Unicode standard was created.
- Objective of Unicode: It unify all the different encoding schemes.

Unicode and UTF

- The Unicode standard defines values for over 128,000 characters and can be seen at the Unicode Consortium.
- Unicode is a character set where as UTF (Unicode Transformation Unit) is an encoding scheme.

UTF : Unicode Transformation Unit

- UTF has several character encoding forms like:
  - UTF-8: Only uses one byte (8 bits) to encode English characters.
  - UTF-16: Uses two bytes (16 bits) to encode the most used characters.
  - UTF-32: Uses four bytes (32 bits) to encode the characters.

Takeaways

- Types of text data.
- ASCII Encoding.
- Unicode Encoding using UTF.

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References

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Foundations of Computer Systems

Objectives

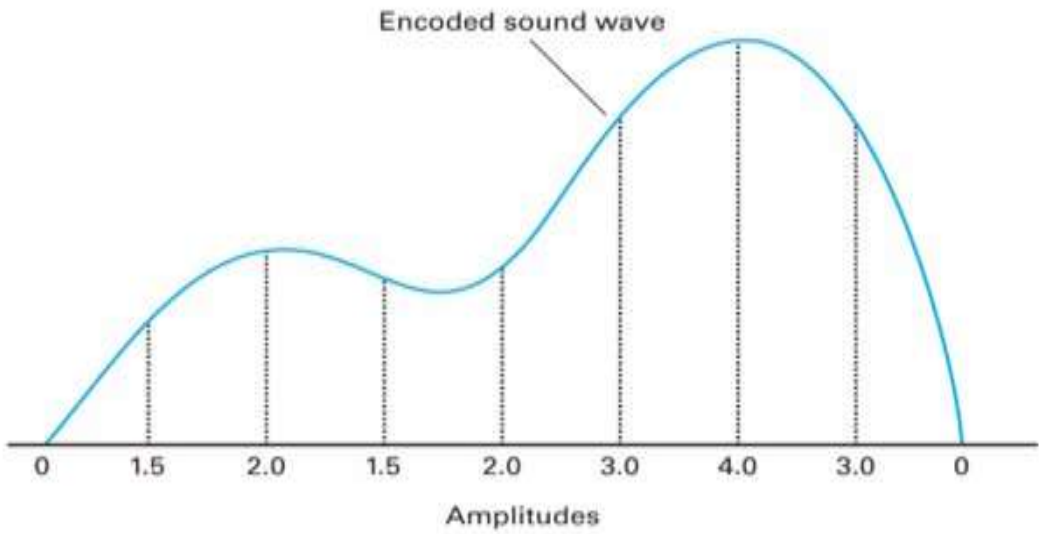
- Analog and Digital Signal.
- Representing Audio as Binary data.
- Digital Image Representation.
- Video Representation.

Audio Representation

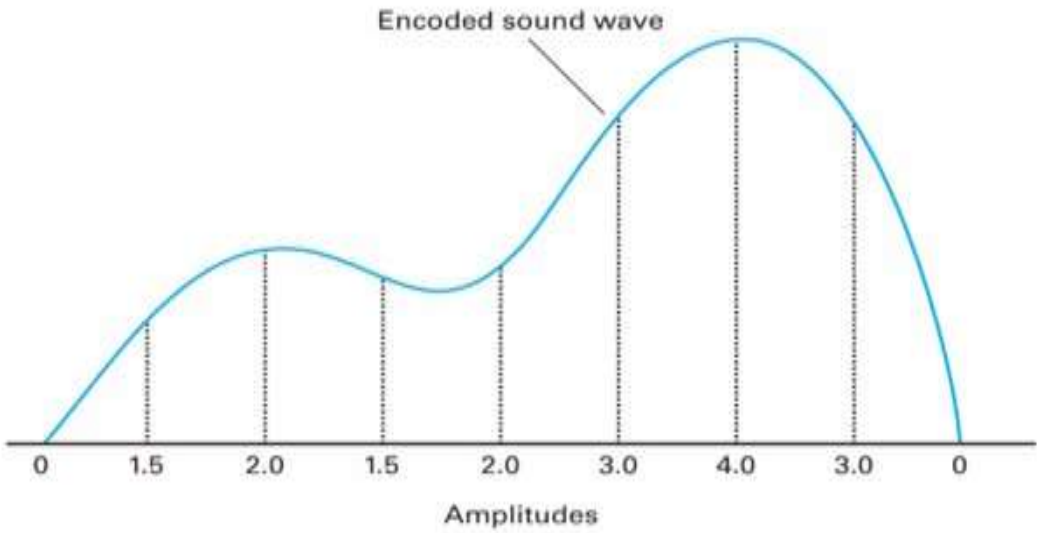
- Analog Signal
  - Continuous & time-varying.
  - Amplitude changes continuously.
  - Often represented in the form of sine waves.
  - Audio data is analog in nature.
- Digital Signal
  - Discrete and time separated
  - Represented by square waves

Representing Audio as Binary Data

- Quantization :- The amplitude of analog audio data, which has a continuous range of values, is fir
discrete values.
- Sampling:- The samples whose ar
is then obtained.
- The amplitude of the samples are its corresponding binary values.



Representing Audio as Binary Data



Digital Image Representation

- Image: A collection of picture elements called as pixels, arranged as rows and columns of a matrix.
- Resolution of the image: The number of pixels used to represent an image.
- Each pixel has its own intensity value, or brightness.
- Image type varies according to the type of intensity used for each pixel.

Types of Images

- Binary Image
- Grayscale Image
- Colour Image

Binary Images

- It contains only two colours – Black and White.
- Black is represented as 0 and White as 1.
- Each pixel can be stored using a single bit.

Grayscale Images

- A range of shades from the black to white colour.
- Uses 8-bit format that accommodates 256 different shades.
- The range of shades varies from 0-255,
  - 0 for Black
  - 255 for White

Colour Images

- Each pixel is composed of 3 channels:
  - **R**ed, **G**reen and **B**lue colours.
- Each of these channels need 8 bits for storage and hence a total of 24 bits is required to represent a single pixel.
- The shade of each pixel will vary depending on the intensity of R, G or B.
- For example:
  - (0,0, 255) is completely blue color.
  - (255,0,0) is completely red.
  - (255,127,0) is orange in color.

Digital Video

- A series of digital images displayed in rapid succession.
- In the context of video these images are called frames.
- The rate at which frames are displayed is known as the frame rate and is measured in frames per second (FPS).
- Representing a video is nothing but representing these frames in an efficient manner.

Takeaways

- Audio Representation.
- Digital Image Representation.
- Types of images.
- Video Representation.

References

[1] Fundamentals Computers, V Rajaraman, Neeharika Adabala, Phi Learning Pvt. Ltd

[2] J. Glenn Brookshear,"Computer Science: An Overview", Addision-Wesley, Twelth Edition, 2014

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